

GSLC Session 4 - Machine Learning

Individual Task

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Classifications Topics

1. What type of classification is the case below:

a) Apakah hari ini akan hujan? (Binary Classification)

x_1	x_2	x_3	x_4	Rain (y)
<i>value</i>	<i>value</i>	<i>value</i>	<i>value</i>	Yes/no

b) Ada buah apa saja didalam foto tersebut? (Multilabel Classification)



x_1	Grape (y_1)	Manggo (y_2)	Kiwi (y_3)	Apple (y_4)	Watermelon (y_5)	Banana (y_6)
<i>photos</i>	Yes/no	Yes/no	Yes/no	Yes/no	Yes/no	Yes/no

c) What kind of house suits my finances and how to pay for it? (Multi Output Classification)

x_1	x_2	x_3	x_4	House Type (y_1)	Payment method (y_2)
<i>value</i>	<i>value</i>	<i>value</i>	<i>value</i>	Type A/ Type B/ Type C/ Type D	Credit/Cash

d) What kind of sofa suits my living room? (Multiclass Classification)

x_1	x_2	x_3	x_4	Sofa (y)
<i>value</i>	<i>value</i>	<i>value</i>	<i>value</i>	Type A/ Type B/ Type C/ Type D

e) What should I wear today? (Multioutput Classification)

x_1	x_2	x_3	x_4	Top (y_1)	Pants (y_2)	Shoes (y_3)
<i>a value</i>	<i>a value</i>	<i>a value</i>	<i>a value</i>	t-shirts/ shirts/ suit/ coat	Jeans/ Cargo pants/ Slim-fit pants/ Tech pants	Combat boot/ Loafer/ Oxford shoe/ Sports shoes

2. Make examples of each binary, multi-class, multi-label, and multi-output classification, such as question number one, accompanied by questions, tables, appropriate variable names for features(x) and labels (y), and possible answers for each target variable/labels (y).

Binary : Membership Classification

Am I a Member?

Payment Status (x_1)	Membersip Status (y_1)
	Member/Non-Member

Multi-Class : Air Pollution Classification

What is today air quality?

CO2 Rate (x_1)	CO Rate (x_2)	O2 Rate (x_3)	N Rate (x_4)	Other Particle Rate (x_5)	Air Quality Index
					Good / Moderate/ Unhealthy / Hazardous

Multi-Label : Weather Classification

What is todays weather status?

Temperature (x_1)	Humidity (x_2)	Air-Speed (x_3)	Sunny (y_1)	Rainy (y_2)	Cloudy (y_3)
			Yes / No	Yes / No	Yes / No

Multi-Output : Routine Classification

What should I do today?

Energy Levels (x_1)	Motivation Levels (x_2)	Health Levels (x_3)	Morning (y_1)	Noon (y_2)	Afternoon (y_3)
			Jogging / Gym / Morning Coffee / Sleep Again	Work / Study / Gym / Sleep Again	Sleep / Study / Work / Watch K-Dramas / Play Games

3. Guess the No Information Rate (NIR) value of this case:

Case1: I have a train and test set containing balanced data like this: Train set: Grape = 25 pictures Orange = 25 pictures Melon = 25 pictures Banana = 25 pictures Test set: Grape = 5 pictures Orange = 5 pictures Melon = 5 pictures Banana = 5 pictures	Case2: I have a train and test set containing unbalanced data like this: Train set: Grape = 70 pictures Orange = 10 pictures Melon = 10 pictures Banana = 10 pictures Test set: Grape = 14 pictures Orange = 2 pictures Melon = 2 pictures Banana = 2 pictures
Questions	
NIR = 25%	NIR = 70%
If I train a model with accuracy 60%. Is my model better than the NIR? yes	If I train a model with accuracy 60%. Is my model better than the NIR? no

* Clue see ppt session 3 or 4 in slide 26

4. Explain the differences between **Precision** and **Recall**. Why is it different? What is the purpose of using Precision and Recall?

The difference between Precision and Recall is

Precision is the proportion of positive value that were identified correctly by computing the ration of True Positive over All Real Positive Value, and because of it we can see how useful a model is. Precision is suitable to be used in customer prediction system where it is better to raise false negative than to be outright wrong.

Recall is the proportion of the true positive value that were identified by computing the ration of True Positive with all true positive value. It is important because it enables us to see how complete a model is. Recall is suitable in medical model because it is better to raise false positive than to let someone undiagnosed.

5. What is **F1 Score**? And explain when is it the right time to use the F1 score?



F1 Score is a evaluation metric that's like a combination between precision and recall because of it's effectiveness if

- False Positive and False Negative is equally costly.
- Adding more data is doesn't change the outcome.
- High rate of False Negative

F1 Score example use case is to evaluate search performance metrics where getting False value is not a good thing and ideally able to get the true value every time.

Regression Topics

6. Explain the differences between **Linear Regression** and **Polynomial Regression**.

Linear Regression is a regression that makes a linear equation from the independent variables and intercept to the dependent variable. Such that it assumes the dependent variable can be explained by a linear equation and can only capture the linear function of the dependent variable. When plotted the linear regression will only make a straight line.

Polynomial Regression is a regression that can't just capture the linear function of the dependent variable but also able to capture the nonlinear function of the dependent variable as well. Thus it doesn't assume that the dependent variable is a linear function but can also be non-linear. When plotted Polynomial Regression can make a curve line.

7. Explain the differences between **underfitting** and **overfitting**.

Overfitting is a term for when a model is too right for the train data such that it performs only good on the train data but poorly when testing is done.

Underfitting is a term for when a model is just bad at predicting the dependent variable such that it will perform poorly in both train and test data.

8. coefficient of determination (R^2) value is always between 0 and 1. Explain what does it imply?

The coefficient of determination or R^2 is the coefficient for the correlation between the observed value and the predicted value squared. Basically, the closer it is to one implies that the model is well fit, and the more it is closer to 0 implies that the model is not well fitted, which could be because of wrong algorithm and/or high error variance.

9. Can the coefficient of determination (R^2) be below zero? In practice, sometimes the (R^2) achieved below zero, then why does it happen and what does it mean?

The R^2 in general can't be negative because it is the correlation coefficient of the observed value and predicted value squared. But, in some cases the R^2 can be negative when the model is worse at predicting than just using the mean of the dependent variable. Negative R^2 means that the model used is not appropriate with the data given.

10. We can use Pearson Correlation Coefficient (PCC) to see the correlation between two variables.

The PCC value is always between -1 and 1. Explain what does it imply?

PCC value quantifies the linear relationship between 2 variables. PCC value closer to 1 implies that the 2 variables are positively correlated. Meaning 1 PCC value means that variable a and b are perfectly positively correlated. Meanwhile PCC value closer to -1 implies that the 2 variables are negatively correlated. Meaning -1 PCC value means that variable a and b are perfectly negatively correlated. And PCC of 0 means that variable a and b are not correlated at all.

AOL Topics

11. Mention the names and NIMs of your AOL group members?

- Elvina Benedicta Santoso - 2602112322
- Wilbert Chandra -2602113666
- Felix Cristiano Bungaran - 2602170612
- Jovan Samuel Hendra - 2602167832
- Kayla Masayuningtyas - 2602141871
- Meisa Kamilia - 2602135446

12. What type of Machine Learning Project will you perform as AOL project? Classification

13. Explain the problems you will raise in your AOL assignment?

In our AOL assignment, we will be addressing the healthcare theme. As we are aware, heart disease is one of the most deadly diseases globally. Therefore, we are interested in creating a machine learning model to predict the presence or absence of cardiovascular disease. Based on the Binus Sustainable Development Goals (SDG), the suitable SDG topic for our case is "Good Health and Well-being."

14. Based on the Binus Sustainable Development Goals (SDG), What are the SDG topics that are suitable for your case?

The SDG topics that are suitable for our case is the number 3, **Good Health and Well-being**

